Computer Vision Lab Mid Exam: HOG vs SIFT on BloodMNIST

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Course: Computer Vision Lab (Mid Exam) | **Degree:** BS Artificial Intelligence

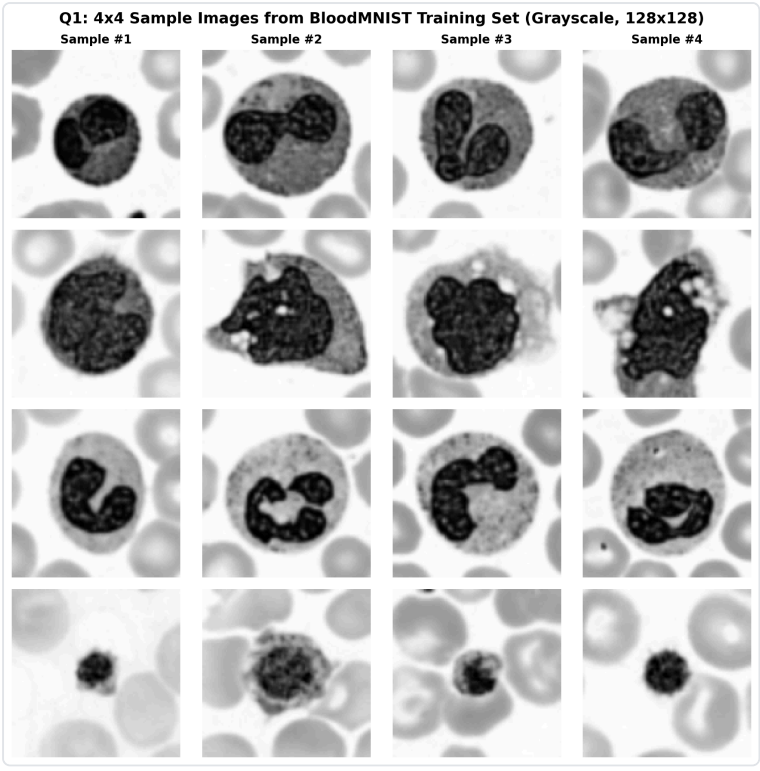
GitHub: https://github.com/Afnan-makstrix/lab-mid-hog-vs-sift

1. Registration Settings Derivation (FA23-BAI-028)

Setting	Derivation Rule	Value for FA23-BAI-028
Random Seed	Last two digits of 028	**28**
HOG Cell Size	Last digit 8 is even → 8x8	**(8, 8)**
SIFT Grid Size	Last digit 8 is in range 5–9 → 6x6	**6 x 6 (36 keypoints)**
Dense SIFT Feature Vector	36 keypoints × 128 dimensions	**4,608**
HOG Feature Vector	225 blocks × 36 features	**8,100**

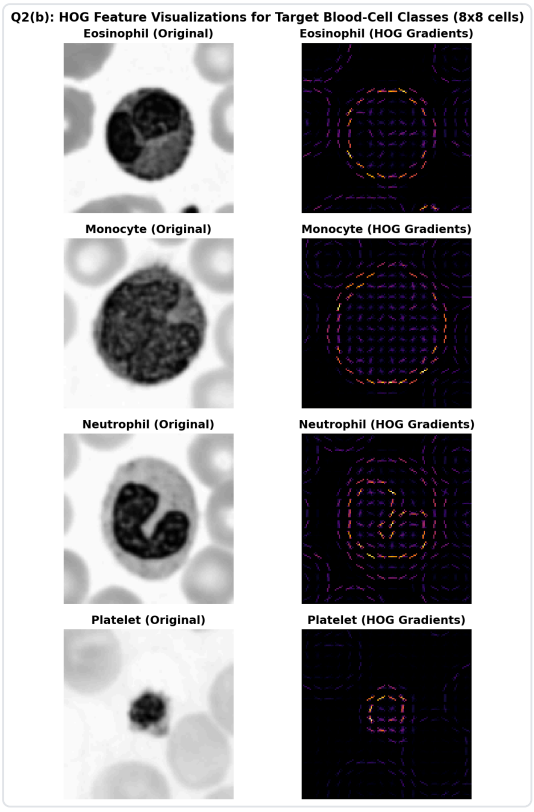
2. Q1 Data Setup (BloodMNIST size=64 → 128x128 Grayscale)

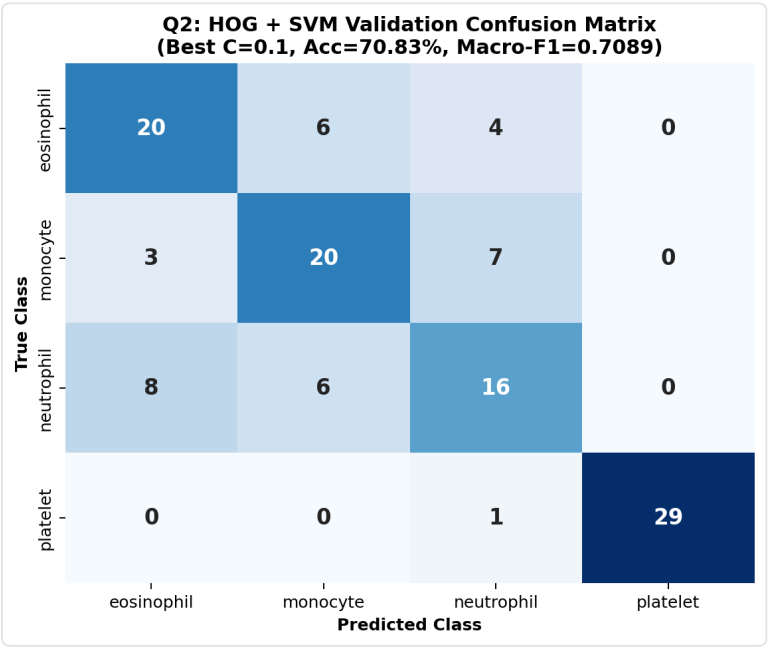
Sampled 100 Train, 30 Validation, 30 Test images per class across 4 classes: `eosinophil (1)`, `monocyte (5)`, `neutrophil (6)`, `platelet (7)`.


Q1: 4x4 Sample Images from BloodMNIST Training Set (Grayscale, 128x128)
Sample #1
Sample #2
Sample #3
Sample #4

3. Q2 HOG Feature Extraction & Classifier

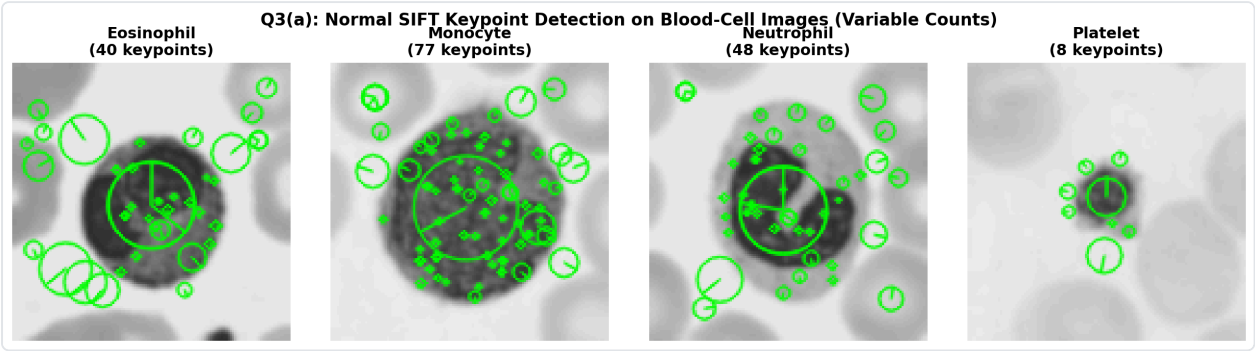
Extracted 8,100-dimensional HOG feature vectors (9 orientations, 8x8 cells, 2x2 block with L2-Hys norm). Tuned LinearSVC with C ∈ (0.1, 1, 10) on validation set. Best model: **C = 0.1** (Accuracy: **70.83%**, Macro-F1: **0.7089**).

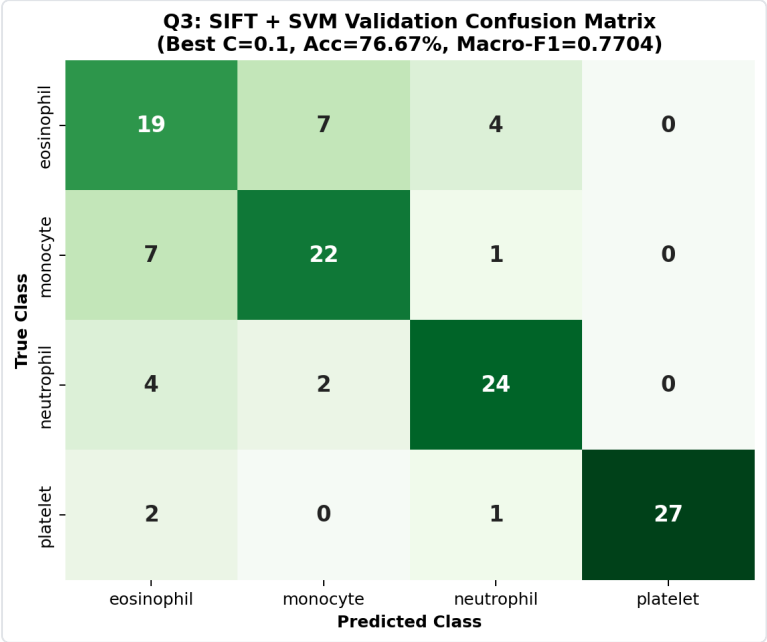




4. Q3 SIFT Feature Extraction & Classifier

Dense SIFT on fixed 6x6 grid (36 keypoints, scale=32) yielding **4,608 dimensions**. Tuned LinearSVC with C ∈ (0.1, 1, 10) on validation set. Best model: **C = 0.1** (Accuracy: **76.67%**, Macro-F1: **0.7704**).

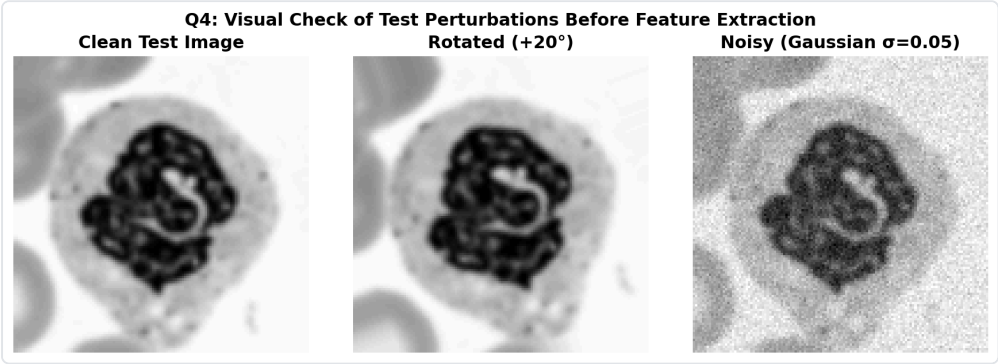


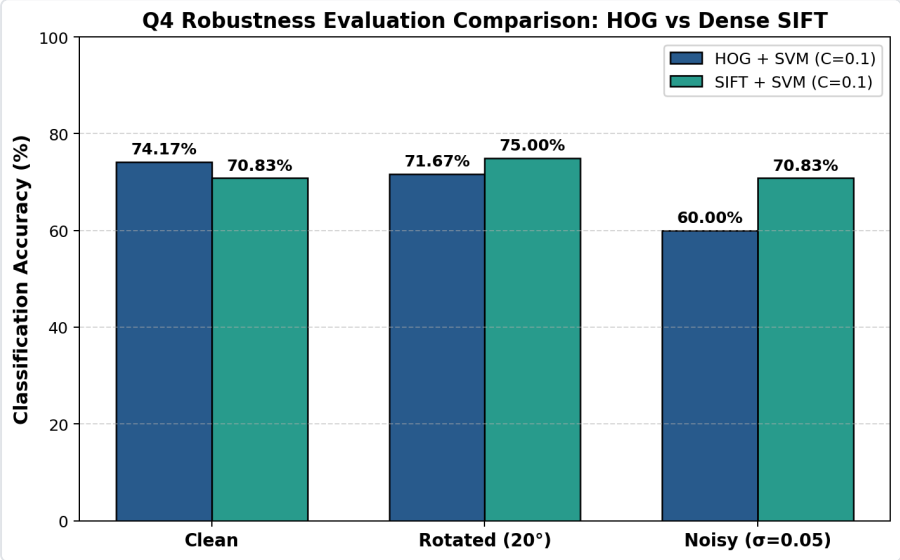


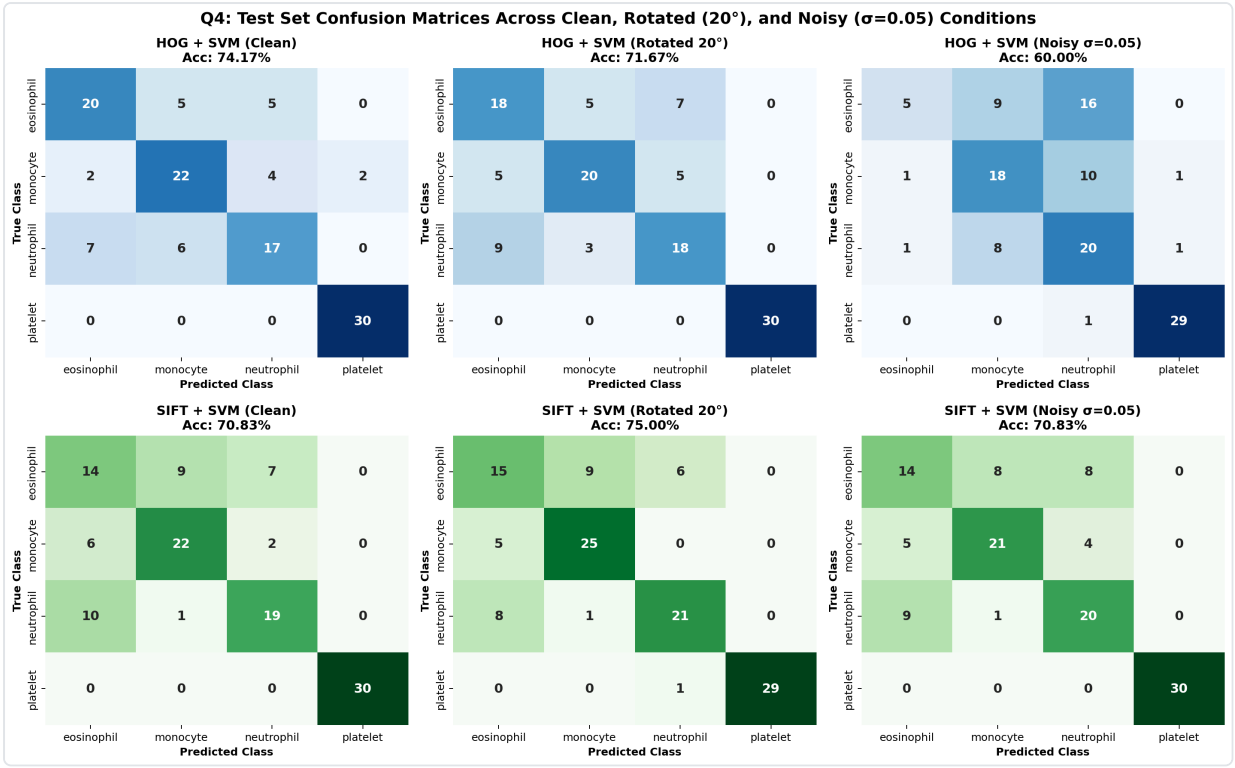
5. Q4 Robustness Evaluation on Test Set

Frozen models evaluated under 3 conditions: Clean, Rotated (20°), and Noisy (Gaussian σ=0.05 on [0, 1]).

Model	Clean Accuracy	Rotated (20°) Accuracy	Noisy (σ=0.05) Accuracy
HOG + LinearSVC (C=0.1)	**74.17%**	71.67% (-2.50%)	60.00% (-14.17%)
SIFT + LinearSVC (C=0.1)	**70.83%**	**75.00% (+4.17%)**	**70.83% (0.00%)**






Q4: Test Set Confusion Matrices Across Clean, Rotated (20°), and Noisy (σ=0.05) Conditions
HOG + SVM (Clean)
Acc: 74.17%
HOG + SVM (Rotated 20°)
Acc: 71.67%
HOG + SVM (Noisy σ=0.05)
Acc: 60.00%
SIFT + SVM (Clean)
Acc: 70.83%
SIFT + SVM (Rotated 20°)
Acc: 75.00%
SIFT + SVM (Noisy σ=0.05)
Acc: 70.83%
True Class
Predicted Class
eosinophil
monocyte
neutrophil
platelet

6. Q5 Conceptual Questions & AI Reflection

Q5(1)(a) Block Normalisation in HOG

Provides invariance to illumination, shadows, and contrast changes by dividing block histogram vectors by their L2 norm, cancelling uniform multiplicative lighting changes while preserving relative directional edges.

Q5(1)(b) Hyperparameter Tuning on Test Set

Tuning C on the test set causes data leakage, resulting in an overly optimistic biased score rather than an honest measurement of generalization on unseen clinical samples.

Q5(2) Most Confused Pair: Eosinophil vs Neutrophil

Both cells are ~12–15 µm granulocytes with segmented lobed nuclei, making them look identical in single-channel grayscale. In Wright-Giemsa histology, eosinophils are distinguished by bright red/orange granules and neutrophils by pale pink granules; converting to grayscale discards this essential chromatic signature.